Kiarash Rezaei

Doctoral Researcher | Chalmers University of Technology

Gothenburg, Sweden | kiarashr@chalmers.se | Office: +46 31 772 28 86
kiarashrezaei.github.io/kiarash-rezaei | Chalmers Profile | GitHub | LinkedIn | Google Scholar | ORCID

Research Interests

Trustworthy autonomous networks and O-RAN; uncertainty quantification, conformal prediction and risk control; reliable decision-making under distribution shift.

Education

PhD in Electrical Engineering (in progress) 2025 – Present
Chalmers University of Technology Gothenburg, Sweden
Department of Electrical Engineering. Supervisor: Prof. Paolo Monti; co-supervisor: Dr. Carlos Natalino.

MSc in Telecommunication Engineering 2021 – 2024
Politecnico di Milano Milan, Italy
Signal and Data Analysis track. Grade: 106/110.
Thesis: *Continuous IMU-MEMS Self-Calibration Process by Means of Tiny Neural Networks.*
Supervisor: Prof. Luca Barletta.

BSc in Computer Science 2014 – 2020
Kharazmi University Tehran, Iran
Department of Mathematics and Computer Science. Supervisor: Prof. Keivan Borna.
Thesis: *Automatic Architecture Design of CNNs using Genetic Algorithm and Reinforcement Learning (MetaQNN).*

Research Experience

Doctoral Researcher Feb 2025 – Present
Chalmers University of Technology Gothenburg, Sweden
Communication, Antennas and Optical Networks (CAOS) Division, Department of Electrical Engineering.

- Led development of **TRACE-CRC** for whole-trajectory uncertainty calibration in multi-step CSI prediction, obtaining tighter uncertainty regions than conservative multi-step baselines [C7].
- Led design and evaluation of **Conformal QoT**, translating calibrated predictions into lightpath-feasibility decisions, improving accuracy from 92% to 99.6% on evaluated open datasets [C4].
- Led development and expert evaluation of LLM-based explanations incorporating SHAP feature interactions; expert evaluation reported 12.2% greater usefulness than a SHAP-only prompt baseline and 97.5% correctness [C3].

AI Researcher Nov 2023 – Jul 2024
STMicroelectronics Milan, Italy

- Designed an end-to-end pipeline for continuous self-calibration of MEMS inertial sensors, integrating tiny neural networks with intelligent sensor processing units.
- Applied quantization-aware training to memory-efficient neural networks for sensor self-calibration on resource-constrained hardware [J1], [C2].

Computer Vision Intern Oct – Nov 2023
Envision The Hague, Netherlands

- Benchmarked multimodal models trained on egocentric datasets across multiple evaluation metrics.

Machine Learning Researcher Apr 2022 – Jul 2023
PoliMi Data Science Association (PMDS) Milan, Italy

- Collaborated with Sares Miramondi on production-line anomaly detection; analyzed public sentiment toward COVID-19 in a separate sentiment-analysis project.

Publications

Journal Articles and Submitted Manuscripts

[J1] Inertial Measurement Unit Self-Calibration by Quantization-Aware and Memory-Parsimonious Neural Networks
Cardoni, M., Pau, D. P., **Rezaei, K.**, & Mura, C.
Electronics, 13(21), 4278, 2024. DOI

[J2] Conformal Risk Control for QoT-Aware Lightpath Feasibility Under Technology-Driven Domain Shift
Rezaei, K., Ayoub, O., Monti, P., & Natalino, C.
Submitted to *Journal of Lightwave Technology (JLT)*.

[J3] Reliable Pilot Scheduling for Multi-Step CSI Prediction with Calibrated Epistemic Uncertainty
Rezaei, K., Sattari, M., Aliakbari, J., Svensson, T., Monti, P., & Natalino, C.
Submitted to *IEEE Wireless Communications Letters (WCL)*.

Conference and Workshop Papers

[C7] TRACE-CRC: Trajectory-Adaptive Conformal Risk Control for Multi-Step Channel State Information Prediction
Rezaei, K., Sattari, M., Aliakbari, J., Svensson, T., Monti, P., & Natalino, C.
15th Symposium on Conformal and Probabilistic Prediction with Applications (COPA), PMLR 329, pp. 984–1009, 2026. Paper | Code

[C6] Human-Grounded Evaluation of Large Language Models for Optical Network Automation
Rezaei, K., Ayoub, O., Monti, P., & Natalino, C.
26th International Conference on Transparent Optical Networks (ICTON), pp. 1–6, 2026. Invited paper. DOI

[C5] Bridging the Trust Gap in AI-Driven Optical Networks with Structured Explainability
Rezaei, K., Ayoub, O., Natalino, C., & Monti, P.
Opto-Electronics and Communications Conference (OECC), 2026. Invited paper. Record

[C4] Policy-driven Conformal Prediction for Trustworthy QoT Estimation
Rezaei, K., Ayoub, O., Monti, P., & Natalino, C.
Optical Fiber Communication Conference (OFC), 2026. DOI

[C3] Generative Explainability for Next-Generation Networks: LLM-Augmented XAI with Mutual Feature Interactions
Rezaei, K., Ayoub, O., Troia, S., Lelli, F., Monti, P., & Natalino, C.
GenXNet Workshop, IEEE International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob), 2025. DOI

[C2] IMU User Transparent Tiny Neural Self-Calibration
Cardoni, M., Pau, D. P., & **Rezaei, K.**
IEEE Forum on Research and Technologies for Society and Industry Innovation (RTSI), pp. 619–624, 2024. DOI

[C1] An Introduction to Convolutional Neural Networks & Applications
Rezaei, K. & Zamani, S.
National Conference on Contemporary Issues in Computer and Information Sciences (CICIS), 2019.

Teaching and Supervision

Bachelor's Thesis Supervisor 2026 (completed)
Chalmers University of Technology Gothenburg, Sweden

- Supervised the EENX16 thesis *Autonomous Knowledge Agent: From Query to Report without Human Guidance*, on autonomous AI systems.

Teaching Assistant Aug 2025 – Present
Chalmers University of Technology Gothenburg, Sweden
EEN060/EEN065: Applied Object-Oriented Programming.

- Design examinations and lead Python exercise and laboratory sessions on object-oriented programming and practical software implementation.

Industry Experience

Generative AI Engineer Oct 2024 – Jan 2025
AGap2 / RINA Milan, Italy

- Developed LLM-based systems to assess the completeness and coherence of inspection reports.
- Integrated models into a digital reporting platform and delivered a functional minimum viable product within three months.

Data Scientist (part-time) Feb 2023 – Feb 2024
DataLobster Paris, France

- Optimized signal-processing and machine-learning algorithms for streaming data to improve performance and reduce latency.
- Developed anomaly-detection models with explainable AI to support root-cause analysis.

Technical Presentation

[T1] Continuous MEMS Self-Calibration Process by Means of Tiny Neural Networks
Cardoni, M., Pau, D. P., & **Rezaei, K.**
STMicroelectronics TechWeek, 2024. Also submitted as an innovation proposal.

Research Expertise and Technical Skills

Reliable ML	Conformal prediction and risk control; uncertainty quantification; calibration for distribution shift and multi-step prediction [C7], [C4].
Network AI	Optical QoT estimation and multi-step CSI prediction; LLM-based explainability and human-grounded evaluation for network automation [C6], [C3].
Edge AI	Quantization-aware training; memory-efficient neural networks; MEMS inertial-sensor self-calibration [J1], [C2].
Programming	Python, MATLAB; PyTorch, TensorFlow, scikit-learn.
Tools	Linux, Docker, AWS, Azure, Git, LaTeX.

Academic Service

Peer Reviewer, *IEEE Networking Letters* 2026
Completed one manuscript review.

Executive Staff, CICIS 2019 2019
National Conference on Contemporary Issues in Computer and Information Sciences.

Professional Membership

IEEE Student Member 2025 – Present

Languages

Persian (native); English (fluent); Italian (elementary).